NephroPatch

Keywords: Kidney – Urea – Sweat – Immobilization – Urease – RTV Silicone Rubber

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Abstract

Chronic kidney disease (CKD) is a growing health concern globally, often remaining undiagnosed until advanced stages (3A). The project aim is to develop a non-invasive monitoring of sweat urea levels tool, combining demographic and historical health data such as age, gender, Medications, and body mass index. The sweat was collected in circular flexible RTV Silicone rubber. The material was treated chemically using piranha solution followed by sodium hydroxide (NaOH) decreasing the contact angle from 112° to 53° being hydrophilic. The sweat was derived from the surface to the collection chambers and became dissociation of urea molecules. *Urease enzyme* was used in the dissociation of urea molecules producing ammonium ions. The measured charges were 9.81 Coulombs with 26.48% deviation from the theoretical estimated value of 12 Coulombs. This immobilized urease was tested in different temperatures, finding that the most suitable one is in the same range of skin temperature (24 °C – 32 °C). The concentration of urea is analyzed using XGBoost ML model trained over 1659 patient - dataset to categorize showing high accuracy in predictions of 97.61% in determining if the patient suffer from CKD or not, and 41% determining the stage of CKD using H1Abc and 38% without Hb1Ac.

Problem

Approximately 850 million people worldwide are estimated to have kidney diseases, most of whom live in low-income and lower-middle-income countries, and a large proportion of these individuals lack access to kidney disease diagnosis, prevention or treatment. **(Jager et al., 2019)**

CKD refers to an irreversible deterioration in renal function that usually develops over a period of years. Initially, it manifests only as a biochemical abnormality but, eventually, loss of the excretory, metabolic and endocrine functions of the kidney leads to clinical symptoms and signs of renal failure. Most patients with slowly progressive disease are asymptomatic until Glomerular Filtration Rate (GFR) falls below 30 mL/min/1.73 m^2 and some can remain asymptomatic with much lower GFR values than this. **(Ralston, 2018)** The overall trend in Kidney Disease Awareness among U.S. Adults with CKD Stages 3–5 is rising with a small rate **(figure 1)**. This is because of its late appearance of symptoms.

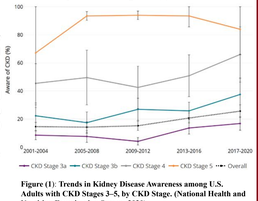


Figure (1): Trends in Kidney Disease Awareness among U.S. Adults with CKD Stages 3–5, by CKD Stage. (National Health and Nutrition Examination Survey, 2020)

- More than 1 in 7 US adults—about 35.5 million people, or 14% are estimated to have CKD.
- As many as 9 in 10 adults with CKD **do not know** they have it.
- About 1 in 3 adults with severe CKD **do not know** they have CKD.

(Chronic Kidney Disease in the United States, 2023, 2024)

Currently, kidney disease is the third fastest-growing cause of death globally and the only non-communicable disease to exhibit a continued rise in age-adjusted mortality. By 2040, CKD is projected to be the 5th highest cause of years of life lost globally. **(Foreman et al., 2018)** This widespread of CKD is because the fact of late diagnoses as the only way to find out if people have CKD is through simple blood and urine tests. The blood test checks for the level of creatinine, a waste product produced by muscles, to see how well the kidneys work. The urine test checks for protein, which may indicate kidney damage. However, these tests require time and money, making routine screening less accessible. As result many CKD cases go undetected until the disease is in its later stages.

Hypothesis

If the current conducted through the electrodes increases after dissociation that is related to an increase in the concentration of urea in sweat, which may correlate with decrease in the renal kidney function state by estimating the glomerular filtration rate (eGFR). Temperature might not cause a significant effect on the system output as its contact with body maintains it. The surface energy of the used material in sweat collection may affect the rate of sweat collection. Data used in training the machine learning model may be affected by a trend in a certain region because of its environment and ethnicity.

Project aim

The project aim is to detect early the chronic kidney disease through estimating the glomerular filtration rate using a wearable device to make it as a routine screening. The goal of the project is to use the urea in sweat as a biomarker to refer to the concentration of the urea in blood to monitor the renal function considering the historical data of the patient. The device will use the urease to react with sweat letting ammonium ions only pass through it and making a connection between the two electrodes and measuring the current. Using the measured current and integrating it with some demographic information about the person through a trained machine learning model which will allows us to make predictions about his renal function state and categorizing the stage of the chronic kidney disease if the patients suffer from it.

Engineering Goals

1. Develop the biosensor to detect at least 10% variation in ammonia concentration after the reaction of the urea with the immobilized urease in the sweat.
2. Design a comfortable sweat collection tool through using hydrophilic and biocompatible material.
3. Using demographic and historical health data from the user of the tool along with urea concentration determination to make accurate predictions of kidney function assessment and determine accurately the stage of the chronic kidney disease through GFR estimation.
4. Ensure long-term usage for at least 30 days of the biosensor through using cost effective and friendly methods for urease immobilization

Methodology

Immobilization of Urease:

- Sodium alginate (80 mg) with 4 ml of urease in 4 ml **(Figure 2)** aqueous medium was thoroughly mixed until a honey-like consistency was attained.
- The mixture was then filled in a syringe and allowed to drop into 50 ml of $CaCl_2$ solution (2 % w/v) from a constant height to form beads.
- The beads were then stirred slowly for 20 min and then removed from the solution and subsequently washed with buffer before use. **(Figure 3)**



Figure (2):Enzyme reagent 4 ml.

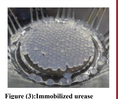


Figure (3):Immobilized urease beads

Assay of immobilized urease:

- The alginate beads with urease entrapped were taken in 1 ml phosphate buffer (0.20 M, pH 7.0) solution and reacted with 1 ml of a 3% & 5% urea solution for 15 min.
- For measurement of the urease activity, the ammonia liberated on incubating the enzyme with urea for a fixed time was determined by measuring the current produced through Arduino circuit board. **(Pithawala et al., 2010) (Figure 4)**



Figure (4): Measuring Circuit

Mold Design & RTV Silicone Housing:

- Two molds were designed to house the silicone using SolidWorks Software, one for the collection from the sweat and the other for the channels and the chamber of urea dissociation.
- The two molds were designed with an internal diameter of 40 mm; the internal height of the collection mold is 7 mm and that for the dissociation 8 mm in height. **(Figure 5)**
- The collection mold contained 6 columns on a circular pattern each of diameter 4 mm and height 7 mm. **(Figure 6)**
- The dissociation mold contained of star shaped 6 columns of height 1 mm connected to the middle chamber of height 5 mm and diameter 10 mm with rectangular cuboid of edges (1 × 1 × 8) mm. These edges are all filleted with a circular fillet of 0.5 mm radius.
- RTV Silicone mixture with the curing agent was housed into the two molds and left for 8 hours. **(Figure 7)**



Figure (5): The collection mold

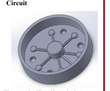


Figure (6): The dissociation mold

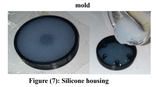


Figure (7): Silicone housing

RTV Silicone Rubber Wet Treatment:

- RTV Silicone Rubber sample is prepared by thorough mixing of silicone raw materials with curing agent in 28.57: 1 (w/w) ratio.
- Wet chemical treatment process for *RTV Silicone Rubber* surface modification, involving piranha solution, mixture of hydrogen peroxide (H_2O_2, 30% purified) and sulfuric acid (H_2SO_4) with the ratio 2: 3 respectively, with the addition of NaOH was the most efficient to convert the surface of the RTV Silicone Rubber to hydrophilic by the decrease of the contact angle. **(figure 8)**



Figure (8):Piranha solution

Training Random Forest Machine Learning Model:

- Collection of a sample of data for 1,659 patients diagnosed with chronic kidney disease (CKD). The dataset includes demographic details, lifestyle factors, medical history, clinical measurements, medication usage, symptoms, quality of life scores, environmental exposures, and health behaviors.
- Preprocessed the dataset by normalizing the values and addressing missing data to ensure it was suitable for training.
- Applied a supervised learning algorithm, such as *Random Forest*, to train the model with eGFR values as the target variable.
- Evaluated the model's performance using metrics like accuracy and mean squared error to validate its predictive accuracy for chronic kidney disease.

Results

Following a series of tests and improvements, positive results have been observed, demonstrating that the prototype meets the engineering goals. These observations include equating the number of charges to the concentration of the urea in solution, the effect of temperature on the immobilized urease, increasing the contact angle of the silicone rubber making it hydrophilic, and the prediction resulted from XGBoost ML Model.

- **Urea Concentration Equating:**

Alginate beads (3 beads) with urease entrapped in 1 ml phosphate buffer (0.10 M, pH 7.5) solution and react with 1 mL of a 3% & 5% urea solution for 15 min. The theoretical amount of charge is estimated using the stoichiometry equation **12.80C** where the molarity of Ammonium ions (NH_4^+) is double the molarity of urea **equation (2).** The difference between the theoretical estimated quantity of charge and the actual values from 1 ml urea solution calculated using **equation (1)** is **9.41C** with **deviation of 26.48%.**

$$Q(T) = \int_{t=0}^{T} I(t).d(t) \qquad (1)$$

- **Hydrophilicity using wet treatment::**

The contact angle of the silicone rubber was decreased from **112°** (hydrophobic) **(figure 9)** to **53°** (hydrophilic) **(figure 10),** to ensure reliable adhesion of sweat and attraction to the chamber.

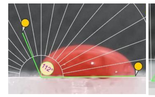


Figure (9): Before treatment (hydrophobic surface)

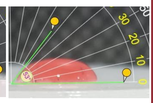


Figure (10): After treatment (hydrophilic surface)

- **Temperature effect on the immobilized Urease:**

Comprehensive testing is conducted to ensure the suitable environment to use the immobilized urease in aim to achieve the highest activity. Alginate beads entrapped in two solutions with equal volumes of 3 mL phosphate buffer and 3 mL urea (3% W/V). These reactions were conducted at different temperatures:

- First range: 24°C - 32°C **(figure 11)**
- Second range: 45°C - 55°C **(figure 12)**
- Third range: 52°C - 65°C **(figure 13)**

Tablet 1): Description of the current resulted from the reaction within different temperature ranges

Temperature (°C) / Current (µA)	24°C - 32°C	45°C - 55°C	52°C - 65°C
Mean	2649.43	786.58	653.14
Min.	2204.30	376.34	0.00
25%	2580.65	645.16	430.11
50%	2634.41	806.45	698.92
75%	2741.94	913.98	913.98
Max.	3010.75	1290.32	1612.90

The statistical summary **(table 1)** revealed that the optimum temperature for the reaction occurred in the range of (24 °C – 32 °C) with average current which is **3.37 times** that was produced in the range of (45°C - 55°C), and **4.06 times** that was produced in the range of (52 °C – 65 °C). The optimum temperature of the urease corresponds to the body's surface temperature ensuring its reliability to be used on skin.

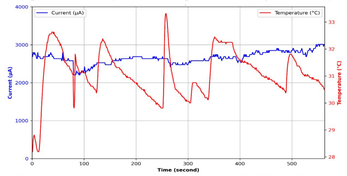


Figure (11):Correlation Between Current and Temperature (24°C - 32°C) Over Time

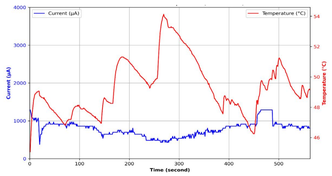


Figure (12):Correlation Between Current and Temperature (45°C - 55°C) Over Time

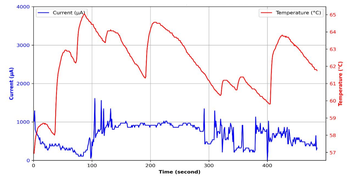


Figure (13): Correlation Between Current and Temperature (52°C - 65°C) Over Time

Machine Learning Model Accuracy:

The XGBoost model trained (using 1659 samples), after data preprocessing of patients and resulted in the following diagnosis:

- Determining (CKD or Non-CKD patient), excluding HbA1c, demonstrates accuracy of 97.61% along with being balanced **as shown in heat map (14)**
- Determining (CKD stage [1 - 5] or non-CKD), excluding HbA1c, demonstrates accuracy of 38.00% along with distribution of predictions **as shown in heat map (15)**
- Determining (CKD stage [1 - 5] or non-CKD), including HbA1c, demonstrates accuracy of 41.00% along with distribution of predictions **as shown in heat map (16)**

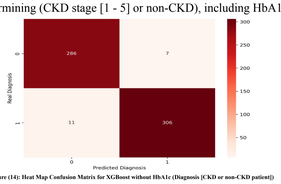


Figure (14): Heat Map Confusion Matrix for XGBoost without HbA1c (Diagnosis [CKD or non-CKD patient])

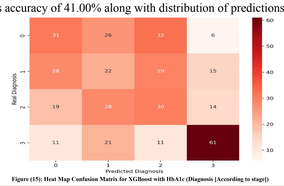


Figure (15): Heat Map Confusion Matrix for XGBoost with HbA1c (Diagnosis [According to stage])

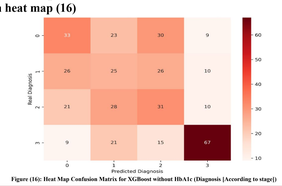


Figure (16): Heat Map Confusion Matrix for XGBoost without HbA1c (Diagnosis [According to stage])

Analysis

Urease Immobilization:

Urease is a nickel-containing enzyme that catalyzes urea to yield ammonia and carbamate, the latter degrades to generate a second molecule of ammonia and carbon dioxide. **(Lubbers et al. 1996 and Khan et al. 2013) (equation 2)**

$$(NH_2)_2CO + H_2O \xrightarrow{Urease} NH_3 + NH_2COOH\ NH_2COOH + H_2O \longrightarrow NH_3 + H_2CO_3 \quad (2)$$

The immobilized enzymes have been used to increase the stability of the bioactive species by protecting the active material from deactivation; reuse significant low-cost operations, easy separation and speedy recovery of the bioactive agent, etc. **(Andrews et al. 1986; Island et al. 1995; Liu et al. 2012; Khan et al. 2013)**

- The optimum temperature of the immobilized enzyme is from 40°C - 50°C
- The optimum pH values for the free and the immobilized enzyme are very close as they are at pH 7.0 and pH 8.0.
- A complete loss in the activity of entrapped enzyme was observed after 11 cycles.
- The decrease in enzyme activity could be due to the leakage of enzyme from the beads, occurring due to the washing of beads at the end of each cycle. **(Elnashar 2010)**

Table (2):Properties distinguishing the immobilized and free urease. (Danial et al., 2015)

Kinetics property	Free Enzyme	Immobilized Enzyme
Optimum pH	7	7-9
Optimum T°	40°C	40-50°C
Thermal T (min)	120	>360
Storage at Room T(day)	<7	>30
Reusability (Cycle)	1	11

RTV Silicone Rubber surface treatment:

RTV Silicone Rubber & Polydimethylsiloxane (PDMS) are members of the siloxane family, derived mainly from silicon, oxygen, and alkane. Currently, oxygen plasma treatment is a common technique used to convert the RTV silicone rubber and PDMS properties from hydrophobic to hydrophilic. However, this technique only renders the surface modified PDMS & RTV Silicone Rubber hydrophilic for a few hours, and the sample had to be used immediately after the treatment. Therefore, in this project, the surface-modification of RTV Silicone Rubber was aimed to be accomplished by using basic chemicals. A two-step chemical treatment was used to treat the RTV Silicone Rubber sample, which is a piranha solution (a mixture of sulfuric acid and hydrogen peroxide) and a potassium hydroxide solution. When sulfuric acid is mixed with hydrogen peroxide, it produces a reactive oxygen atom that reacts with the methyl group in the siloxane polymer. Potassium hydroxide solution then spreads the hydroxyl ions over the surface of the RTV Silicone Rubber sample to create an even hydrophilic RTV Silicone Rubber modified surface. **(Wala et al., 2017)**

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There was set of experiments conducted to show that the contact angle of PDMS (like RTV Silicone Rubber) decreases considerably with the increase of wet chemical treatment time as well as by increasing the concentration of H_2SO_4 in the piranha solution. Moreover, prolonged KOH dip also allows even distribution of hydroxyl ions, thus hydrophilizing the surface uniformly. As a result, the measured contact angle is gradually reduced on increasing the treatment time **(Figure 17).** Increasing the ratio of H_2SO_4 in the piranha solution from 1:1 to 2:3 with the addition of KOH was the most efficient to decrease the contact angle with the surface. **(Maji et al., 2011)** This could achieve a significant decrease in the contact angle of water molecules by 52.68% from 112° to 53°.

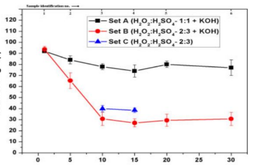


Figure (17):Variation of contact angle vs treatment time for a different set of experimental conditions. (Maji et al., 2011(

Conclusion

To sum up, the lake of awareness about chronic kidney disease led to a significant problem where **9 in 10** with CKD do not know they have it. This resulted as the chronic kidney disease is asymptotic till late stage (3A), where the estimated glomerular filtration rate becomes below **45 mL/min/1.73m²**. So, it was found that the solution for that late detection of CKD is to make a wearable band to collect sweat and use the urea as an indicator to kidney functions. RTV Silicone Rubber material was used in housing the mold of collection tool. Its surface was wet treated using piranha solution followed by NaOH to decrease water's contact angle from **112° to 53°.** The dissociation of urea molecules in the collection chambers is done using immobilized urease to produce ammonium ions (resulting in flowing current). These ions were measured and equated to a **3% & 5% w/v** of urea solution for a volume of 1mL generating **9.81 Coulomb** in a 15-minute experiment. The concentration of data is integrated with the user's infographic and historical health data were analyzed using XGBoost ML model trained over 1659 patient - dataset to categorize if the user suffers from CKD and determine in which stage. The Models could make predictions with accuracy of **97.61%** (Determining CKD or non-CKD patient), **41%** in determining the stage of CKD while HbA1c feature is included, and **38%** in determining the stage excluding HbA1c feature.

Future Plans

- The integration of the sweat collection band with smart watches and digital bands to make it easier for the users to monitor their kidney functions through software programs like (Huawei Health and Samsung Health).
- Training the machine learning model on larger datasets from various countries to ensure having a variety of ethnicities, people with different dietary preferences, lifestyles, and climate adaptations.
- Displacing RTV Silicone Rubber by *Polydimethylsiloxane (PDMS)* material as it offers better biocompatibility and flexibility with microfluidic applications.

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